

HOMEWORK 13

INTRODUCTION TO COMPLEX ANALYSIS

(MATH 4230-40), SPRING 2022

SECTION 5.2 - FRACTIONAL LINEAR AND SCHWARZ-CHRISTOFFEL TRANSFORMATIONS

1. Find fractional linear transformations f satisfying $f(z_i) = w_i$ for $i = 1, 2, 3$ if

(a) $z_1 = -1, z_2 = 1, z_3 = 2$ and $w_1 = 0, w_2 = -1, w_3 = -3$

(b) $z_1 = -1, z_2 = 1, z_3 = 2$ and $w_1 = -3, w_2 = -1, w_3 = 0$

(a) Write $T(z) = w$ for our sought fractional linear transformation T and write $T(z_i) = w_i$ note that Proposition 5.2.4 yields that

$$\begin{aligned} \frac{w - w_1}{w - w_2} \frac{w_3 - w_2}{w_3 - w_1} &= \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1} \\ (w - w_1) \frac{w_3 - w_2}{w_3 - w_1} &= (w - w_2) \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1} \\ w \frac{w_3 - w_2}{w_3 - w_1} - w \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1} &= w_1 \frac{w_3 - w_2}{w_3 - w_1} - w_2 \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1} \\ w \left(\frac{w_3 - w_2}{w_3 - w_1} - \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1} \right) &= w_1 \frac{w_3 - w_2}{w_3 - w_1} - w_2 \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1} \\ w &= \frac{w_1 \frac{w_3 - w_2}{w_3 - w_1} - w_2 \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1}}{\frac{w_3 - w_2}{w_3 - w_1} - \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1}} \end{aligned}$$

This could certainly be simplified further, but there really is no particular benefit to doing so at the moment. Skipping some minor arithmetic in our calculation below, the formula above readily yields our answer:

$$T(z) = \frac{0 \frac{(-3)-(-1)}{(-3)-(0)} - (-1) \frac{z-(-1)}{z-(1)} \frac{(2)-(-1)}{(2)-(-1)}}{\frac{(-3)-(-1)}{(-3)-(0)} - \frac{z-(-1)}{z-(1)} \frac{(2)-(-1)}{(2)-(-1)}} = \frac{z+1}{z-3}$$

Note that we may verify this is correct:

$$T(z_1) = \frac{(-1)+1}{(-1)-3} = 0 \quad T(z_2) = \frac{1+1}{1-3} = -1 \quad T(z_3) = \frac{2+1}{2-3} = -3.$$

(b) Using the same logic as before, we have

$$T(z) = \frac{w_1 \frac{w_3 - w_2}{w_3 - w_1} - w_2 \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1}}{\frac{w_3 - w_2}{w_3 - w_1} - \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1}} = \frac{(-3) \frac{(0)-(-1)}{(0)-(-3)} - (-1) \frac{z-(-1)}{z-(1)} \frac{(2)-(-1)}{(2)-(-1)}}{\frac{(0)-(-1)}{(0)-(-3)} - \frac{z-(-1)}{z-(1)} \frac{(2)-(-1)}{(2)-(-1)}} = z - 2.$$

Note that we may verify this is correct:

$$T(z_1) = (-1) - 2 = -3 \quad T(z_2) = (1) - 2 = -1 \quad T(z_3) = 2 - 2 = 0.$$

2. Prove that any fractional linear transformation T with $c \neq 0$ can be written $T = T_4 \circ T_3 \circ T_2 \circ T_1$, where

$$T_1(z) = z + \frac{d}{c}, \quad T_2(z) = \frac{1}{z}, \quad T_3(z) = \frac{bc - ad}{c^2}z, \quad \text{and} \quad T_4(z) = z + \frac{a}{c}.$$

By means of direct computation, we reach the result after the following manipulation:

$$\begin{aligned} (T_4 \circ T_3 \circ T_2 \circ T_1)(z) &= T_4(T_3(T_2(T_1(z)))) \\ &= T_4\left(T_3\left(T_2\left(z + \frac{d}{c}\right)\right)\right) \\ &= T_4\left(T_3\left(\frac{1}{z + \frac{d}{c}}\right)\right) \\ &= T_4\left(T_3\left(\frac{c}{zc + d}\right)\right) \\ &= T_4\left(\frac{bc - ad}{c^2} \frac{c}{zc + d}\right) \\ &= T_4\left(\frac{bc - ad}{c(zc + d)}\right) \\ &= \frac{bc - ad}{c(zc + d)} + \frac{a}{c} \\ &= \frac{bc - ad}{c(zc + d)} + \frac{a(zc + d)}{c(zc + d)} \\ &= \frac{bc - ad + azc + ad}{c(zc + d)} \\ &= \frac{bc + azc}{c(zc + d)} \\ &= \frac{b + az}{zc + d} \\ &= \frac{az + b}{cz + d}. \end{aligned}$$